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KLEEMANN
KERS and LCM Projects

ENGR 410 Global Engineering Design
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Thank You!

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Thank you for allowing us to work on Project for Kleemann:

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DISCLAIMER

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Recognition for Kleemann and Texas A&M

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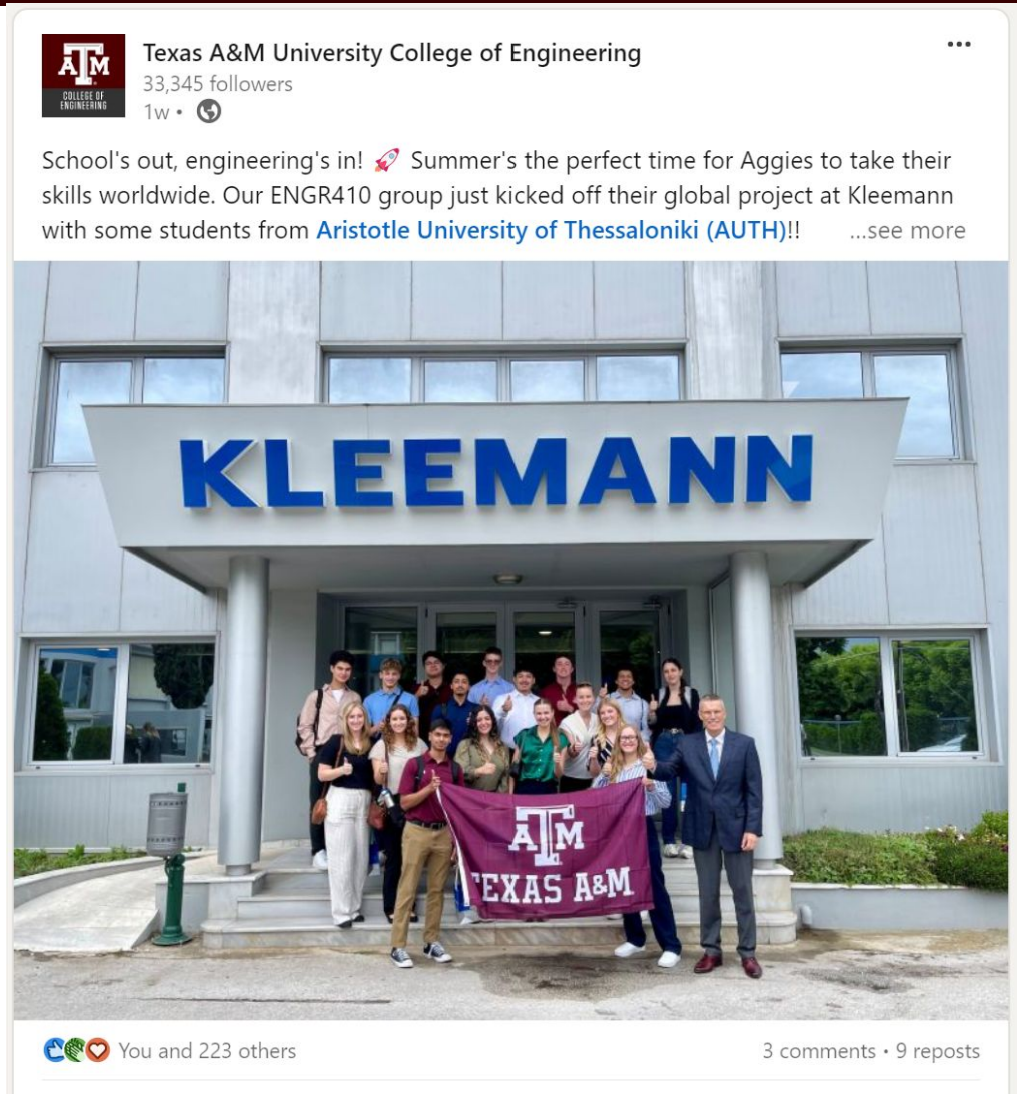
Recognized by **224** stakeholders so far.



Background

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Sustainability Commitment - Greenhouse Gas (GHG) Reduction

KLEEMANN is a well-known elevator manufacturer that provides products and services to several sectors worldwide, such as residential, office, shopping centers, hospitals, hotels, and cruise ships. All stakeholders associated with these industries share the responsibility of reducing GHG emissions.

KLEEMANN is committed to decarbonizing elevator operations, lowering energy costs, improve lifecycle and collaborating with stakeholders on the global decarbonization agenda, which promises significant benefits for GHG reduction.

Background – COP26

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The United Nations (UN) climate change conference in Glasgow (COP26), 2021, brought together 120 world leaders and over 40,000 registered participants, all of whom accepted a cooperation agreement. The agreement focused on all aspects of climate change—the science, the solutions, the political will to act, and clear indications of action.

The world leaders agreed on the following point:

The goal of the agreement is to limit the increase in global average temperature to well below 2°C above pre-industrial levels, with efforts to limit it to 1.5°C. There is deep concern that human activities have already caused around 1.1°C of warming, impacting every region.

Projects Overview

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Focus areas for the overall project challenge was two parts:

- Kinetic Energy Recovery Systems (KERS)
- Life Cycle Management (LCM)

Sustainable Development Goals

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These projects supports SDG 8, 9, 11,12 and 13

Design Thinking Approach

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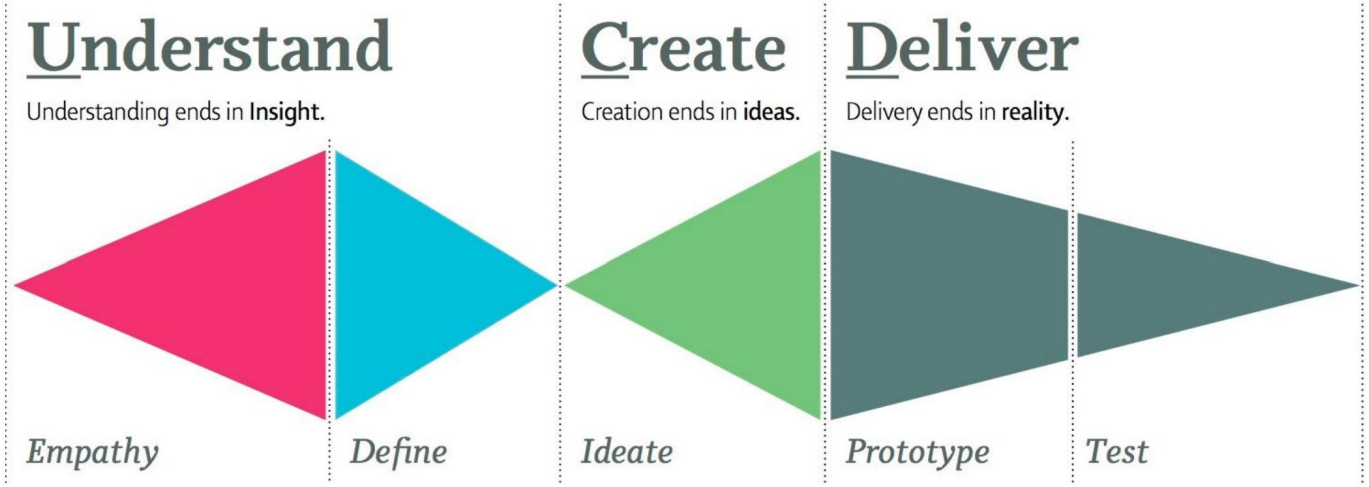

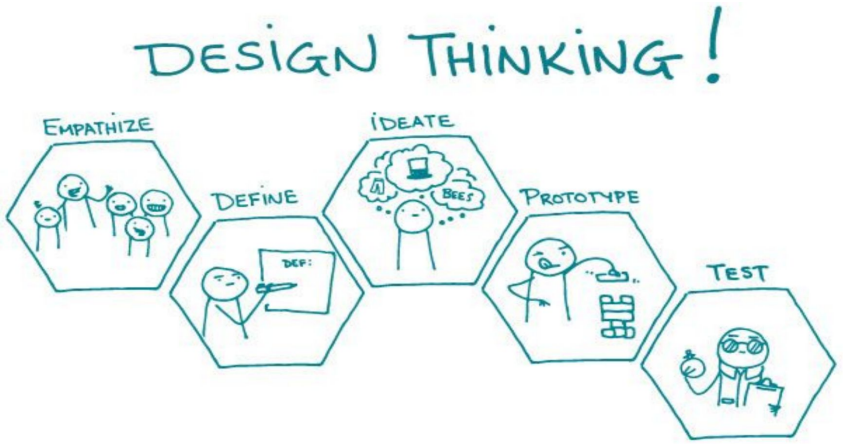

Project - KERS

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Problem Statement: KLEEMANN, an elevator manufacturer, is seeking a strategy to integrate KERS technology into their traction elevators to promote energy efficiency and cost-effectiveness, enhancing vertical transportation

Key Takeaways:

1. Energy efficiency
2. Cost effectiveness

Two Approaches:

1. Harnessing rotational energy of flywheel
2. Electromagnetic design using a coil and magnet

Analytical Process

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Flywheel

$$\Delta \varsigma_T = \frac{1}{2} m(v_f^{\ 2} - v_i^{\ 2})$$

Kinetic Energy Losses

$$\frac{1}{2} m(v_f^{\ 2} - v_i^{\ 2}) = \frac{IVt}{\eta_g}$$

Kinetic Energy to Electrical Energy

$$E_K = \frac{1}{2} I \omega^2$$

Kinetic Energy

MACS

Inputs
Magnet
Material
Magnetic Field Strength (T)
Magnet Radius (m)
Cost ($)
Coil
Material
Wire Thickness (m)
Coil Radius (m)
Number of Turns Per Meter
Cost Per Meter ($)
Resistivity (Ω·m)

Outputs
Power (W)
Voltage (V)
Current (A)
Resistance (Ω)
Wire Length (m)
Cost ($)

$$\varepsilon = -N\frac{BA}{t}$$

Induced EMF (voltage)

Approach #1

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Approach #1: Harnessing rotational energy

Background:

- Main purposes of a flywheel in a car: to balance the engine, store energy to allow more optimization, limit the effects of friction by smoothing the engagement and disengagement in power, and reduce speed fluctuations.
- There are four main types of flywheels: rimmed, solid disc, high velocity, and low velocity.
- Most beneficial: cast iron solid disc
  - High strength and durability with lower maintenance requirements.

Implementation:

- Harness rotational energy into electrical energy and connect flywheel to generator
- Charge a battery to create a flow of electricity back into the elevator or the building

Approach #2

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Approach #2: Electromagnetic design

Courses of Action:

1. A magnet moving through a separate, coil-wrapped tube in coordination with the movement of the elevator
2. Wrap the counterweight shaft in conductive coil and attach magnets to the counterweight
3. A coil extended from the sheave which rotates within a magnetic field

Design #1

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COA #1: A magnet moving through a separate, coil-wrapped tube in coordination with the movement of the elevator

Pros:

- cost effective
- separate system makes it easier to add on to existing elevator systems
- easier to maintain

Cons:

- requires a lot of wire material
- need additional space in elevator shaft
- extra power may be required to move weight of magnet

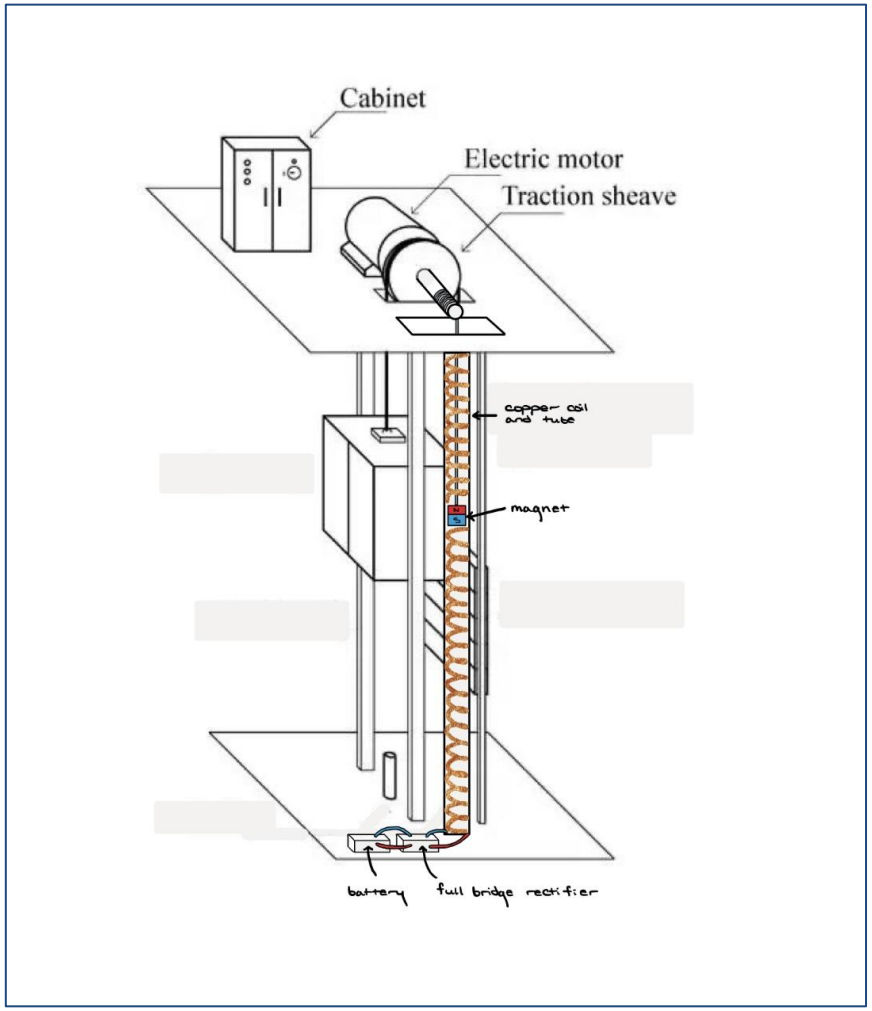

Figure 5: *Technical Diagram of KERS Coil/Magnet System (Design 1)*

Design #2

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COA #2: Wrap the counterweight shaft in conductive coil and attach magnets to the counterweight

Pros:

- magnet weight can be integrated into counterweight
- would not need additional power

Cons:

- requires a lot of wire material
- since it is integrated into elevator, it would not be implemented as an “add-on”

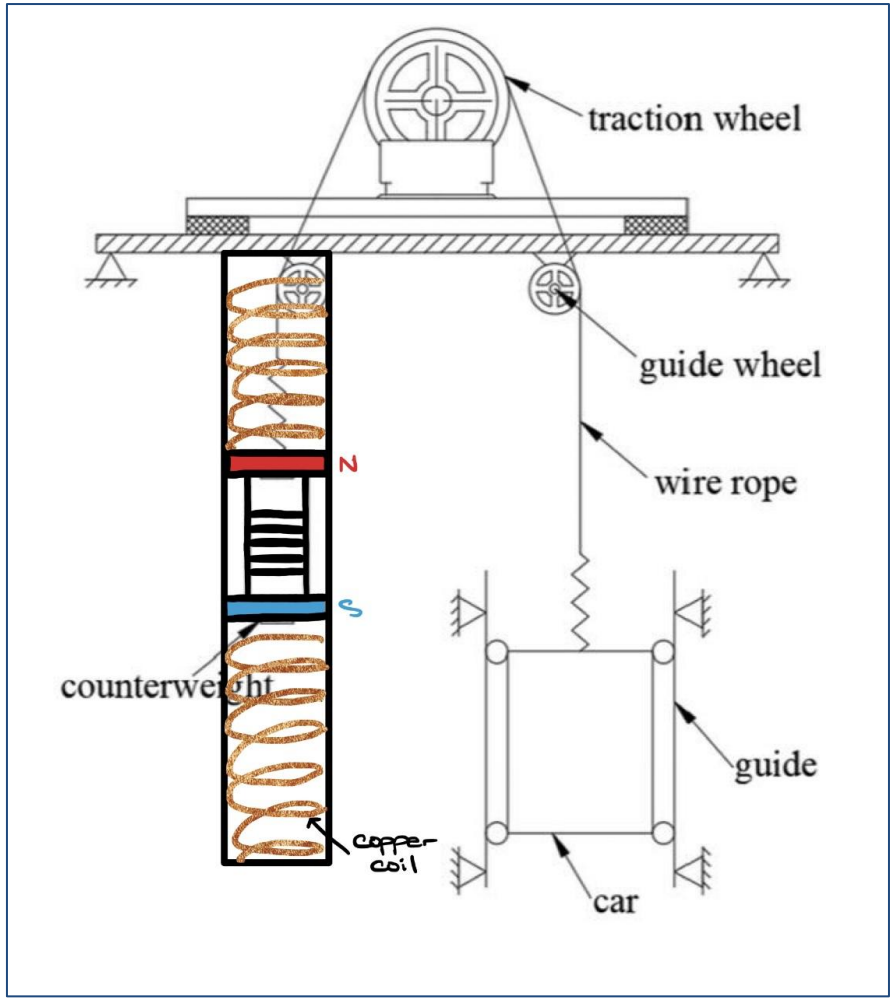

Figure 6: *Technical Diagram of KERS Coil/Magnet System (Design 2)*

Design #3

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COA #3: A coil extended from the sheave which rotates within a magnetic field

Pros:

- less wire material is required
- cost effective
- no need to create separate pulley system

Cons:

- must create spinning shaft off of motor
- would not be implemented as an “add-on,” would be integrated into elevator system
- extra energy may be required to spin coil
- possible resistance added to the spinning sheave (thus resistance to the motor)

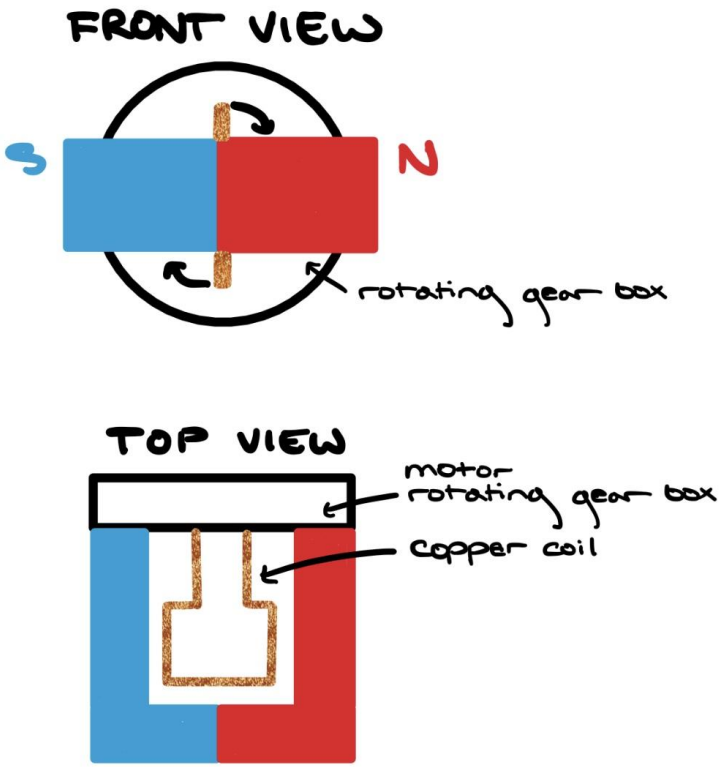

Figure 7: *Technical Diagram of KERS Coil/Magnet System (Design 3)*

Recommendation

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Implementation:

- COA#1 is recommended
- Can be mass produced for any elevator system
  - Not dependent on the actual elevator system
  - Requires space in elevator shaft
- Modifications
  - Short sections of conductive coil instead of entire tube
    - Cost effectiveness
  - Skinnier tube with slimmer magnet to create more turns
    - Energy efficiency
    - Higher voltage and current
  - Stronger magnets produce greater magnetic flux
    - Could increase overall cost of system

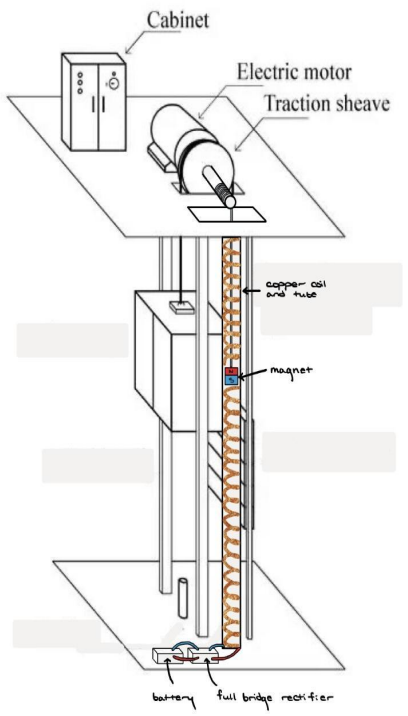

Figure 5: *Technical Diagram of KERS Coil/Magnet System (Design 1)*

Recommendation

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Example System and Parameters:

- COTS Neodymium Magnet, 152.4 mm (6”) OD
- 80/20 Nickel-Chrome Alloy Coil, 3 m long, 2.032 mm OD, 3000 turns
- One coil (1 m/s speed) yields ~1350 W power output
- Five story building could have up to five coils to charge separate capacitors
- $1,700 (cost of the magnet and a single coil, excluding installation costs and electrical storage units)

Problem Statement - LCM

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KLEEMANN aims to improve their products' life cycle by innovating manufacturing processes, minimizing resource use, reducing waste, and lowering carbon footprints, all while maintaining top quality and performance in the elevator industry.

Eco-design Directive

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- Product durability, reusability, upgradability, and repairability
- Energy and resource efficiency
- Recycled content
- Carbon and environmental footprints

Ecodesign Compliant

Initial Thoughts - Sustainability

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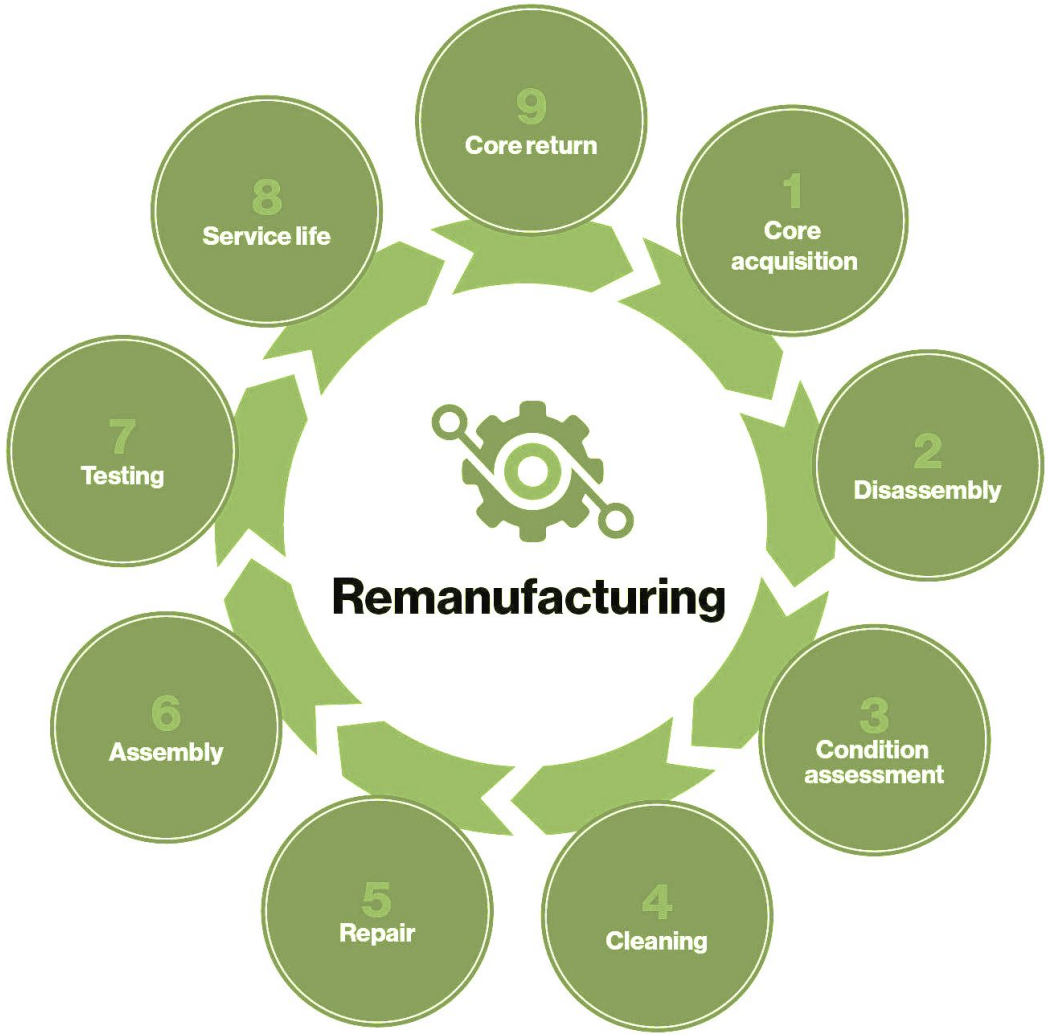

Circular Economy Approach

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Principles of Circular Economy:

- Promotes Sustainable Consumption
- Enhances Product Longevity
- Emphasizes Recycling
- Optimize Manufacturing Processes

Recyclability and Remanufacturing

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Background research:

1. EU funding & Worldwide Policies (Horizon Europe, ISO 9001:2015, EU Lift Directive 2014/33/EU)
2. Elevator standards KLEEMANN follows.
3. Recycling companies (ex. EASA IKE, Global Tech Environmental).
4. Application of circular economy on companies.
5. Disadvantages of recycling facilities ⟹ costly.
6. Which components can be recycled or remanufactured.

Remanufacturing

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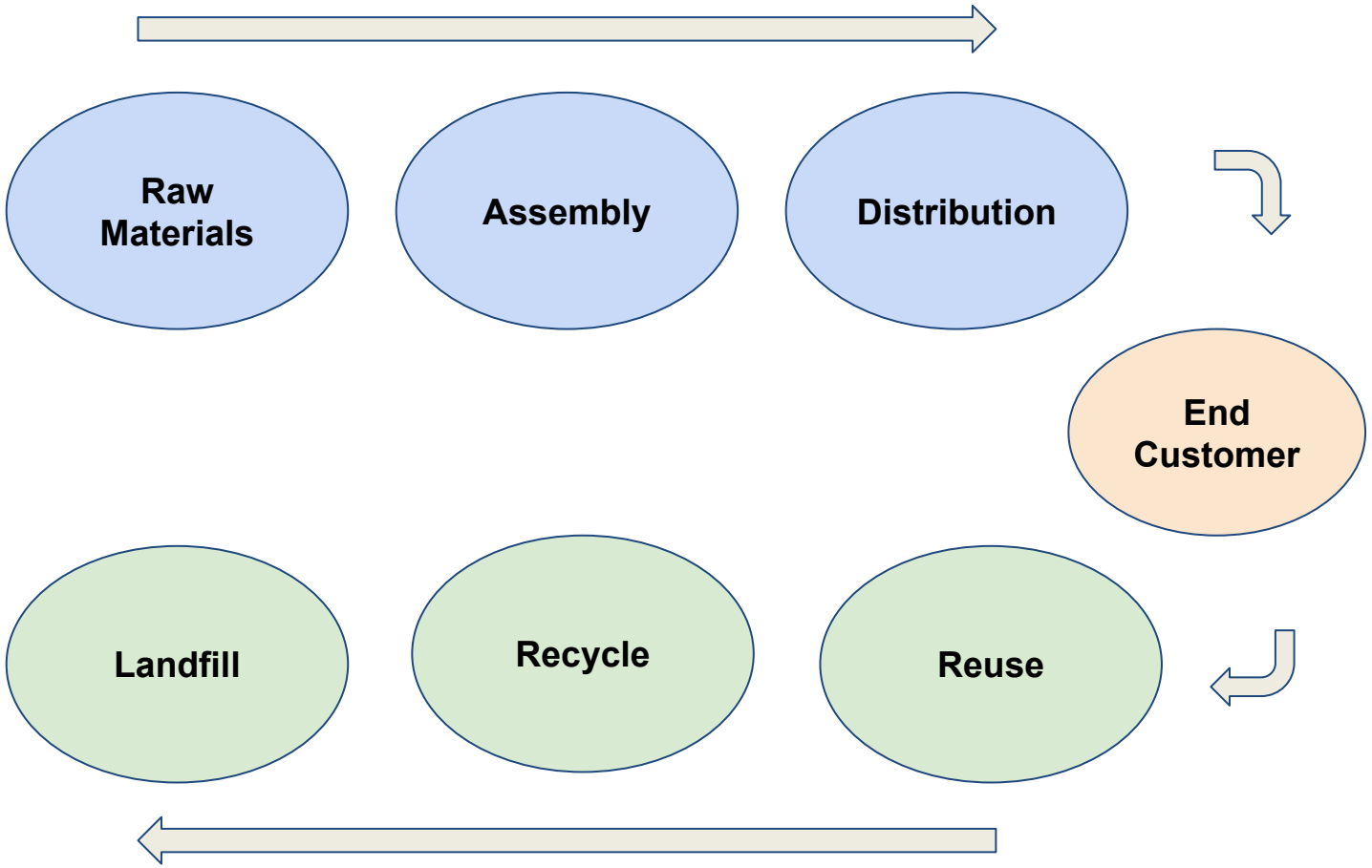
- Definition of Remanufacturing
- Reverse logistics
- Studies & Statistics
- Engaging with Customers and Clients



Recycling

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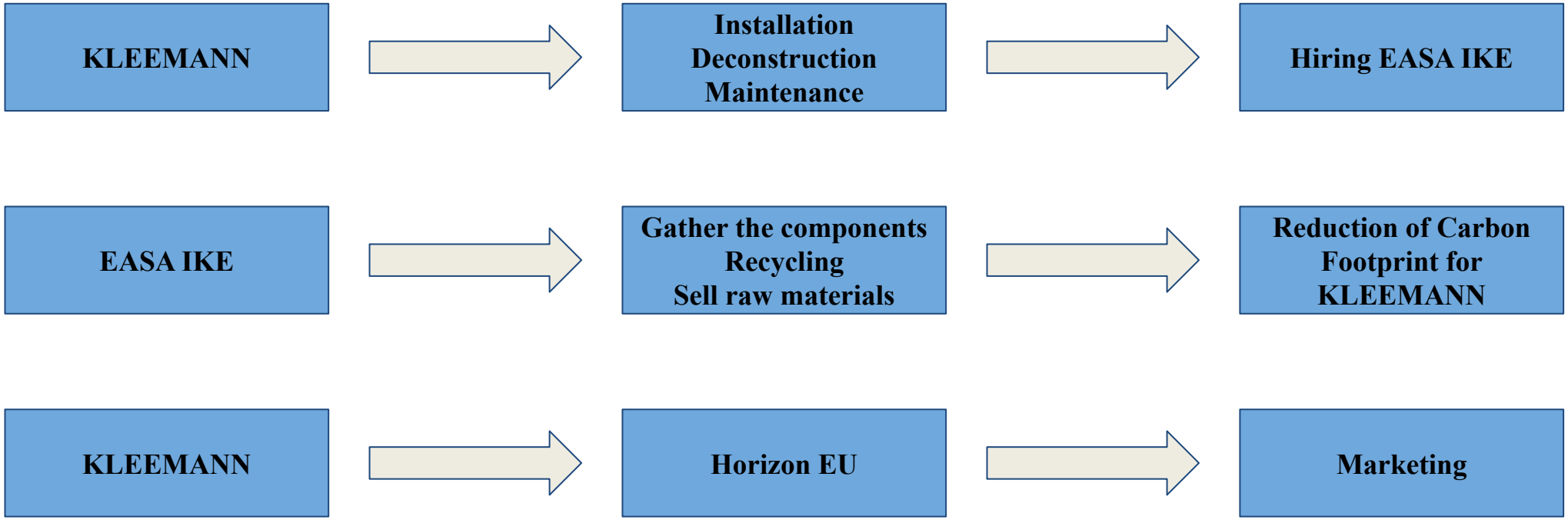
Final Goal: Reduction of Carbon Footprint.



Industry 4.0 - Smart Manufacturing

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Smart Manufacturing / Industry 4.0:

➔ Collection and connection of data

➔ By utilizing IoT systems within the manufacturing process, Kleemann can…

- advance product quality
- improve productivity
- increase production
- manage quality through market expansion
- lessen environmental impact
- manage costs

Smart Manufacturing Challenges

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- ➔ Connecting data across production line
- ➔ Accurate and real-time tracking of data
- ➔ Intelligent decision-making
- ➔ Large investment

Zero Defect Manufacturing (ZDM)

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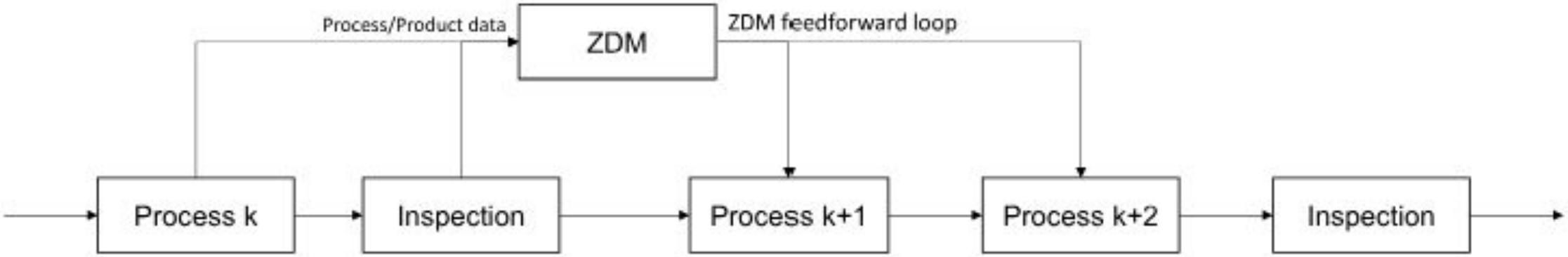
Utilizing all data generated in the production line to recognize and predict potential areas of defect.

Challenges of implementing ZDM for Kleemann:

- High starting cost
- Time consuming
- Meeting ZDM standards



Zero Defect Manufacturing (ZDM)

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Advantages:

- Higher profits
- Material efficiency
- Less waste
- Reduced production time

Learning

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Through this project, students have gained the following knowledge and experience while working on a real world project offered by KLEEMANN:

- Importance of teamwork
- Elevator technology
- Cultural impact
- Research skills
- Communication
- Time management

Questions